CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 20: Area and Perimeter of Plane Figures

Every formula and result of the chapter on one sheet

1	Semi-perimeter	$s = \frac{a+b+c}{2}$
2	Heron's formula	$\sqrt{s(s-a)(s-b)(s-c)}$
3	Equilateral triangle	Area $\frac{\sqrt{3}}{4}a^2$, height $\frac{\sqrt{3}}{2}a$
4	Isosceles triangle	$\frac{b}{4}\sqrt{4a^2-b^2}$
5	Rectangle	Area lb, perimeter $2(l+b)$, diagonal $\sqrt{l^2+b^2}$
6	Square	Area $a^2 = \frac{1}{2}d^2$, perimeter $4a$
7	Parallelogram	Base $\times$ height
8	Rhombus	$\frac{1}{2}d_1d_2$
9	Trapezium	$\frac{1}{2}(a+b)h$
10	Circle	Circumference $2\pi r$, area πr^2
11	Ring	$\pi(R^2-r^2)$
12	Semicircle	Area $\frac{1}{2}\pi r^2$, perimeter $\pi r + 2r$
13	Path outside a rectangle	$(l+2w)(b+2w) - lb$
14	Cross-paths	$w(l+b-w)$